SC301 Numerical Linear Algebra

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1. **Credits:** 3-0-0-3
2. **Prerequisites:** Linear Algebra.
3. **Summary:** A first course in Linear algebra equips students with a set of concepts and tools to handle finite dimensional vector spaces. These are heavily utilized in a wide range of applications in different fields like Image and Signal Processing, Machine Learning, Data Science, Computer Graphics, Mechanical Engineering etc. This course studies algorithms for some core Linear algebra concepts that have been incorporated into libraries like LAPACK, for example, Solving Linear equations and Matrix factorizations like QR , Cholesky and SVD. There are direct and iterative methods for many of these tools.

   These problems in some instances like those encountered in Data Science may involve large matrices which makes it important to study the complexity and stability of these algorithms. The course will examine a wide variety of numerical algorithms, will study them from the perspectives of accuracy and stability, and will discuss a few applications depending on the available time.
4. **Content:** (not necessarily in this order).
   (a) **Review of LA:** Vector Spaces: R^n and C^n, Matrices and Vectors, Linear independence, Rank of a matrix, Subspaces, Eigenvalues and Eigenvectors. Inner products and Norms in R^n and C^n, Orthogonality, Matrix Norms, Spectral Radius.
   (b) **Review of Number representation:** Integer and Floating point representation, Absolute and relative errors.
   (c) **Linear Equations:** Applications: Numerical Solution of PDEs, Curve fitting (special case of Least squares), Gaussian elimination, LU decomposition, Partial and Full pivoting, Stability and efficiency – Condition number. Special cases – Tridiagonal systems, SPD matrices. Iterative Methods: Order of Convergence, Jacobi, Gauss-Seidel, Successive over-relaxation(SOR), Steepest Descent, Conjugate Gradient. Preconditioning.
   (d) **Matrix Decomposition - QR and Cholesky:** Applications – Computing determinant, Orthonormal basis from a basis, Methods: Gram Schmidt Orthogonalization, Givens rotations, Householder reflections. Cholesky Decomposition.
   (e) **Eigendecomposition:** Symmetric and Unsymmetric Eigenvalue problem. Applications – Difference and Differential equations, Method: Power iteration, QR iterations, Hessenberg and Schur forms.
   (f) **SVD:** (if time permits) - SVD – Existence, Interpretation, Applications – Least squares, PCA.

5. **References:**
   (a) Matrix Computations, Golub and Van Loan, John Hopkins Press, 4th ed., 2013
   (b) Numerical Linear Algebra with Applications, William Ford, Academic Press, 2015.
   (c) Numerical Linear Algebra, Trefethen and Bau III, SIAM, 1997.

6. **Evaluation:** (Assuming $\leq$ 50 students)
   (a) Quizzes: 20%
   (b) In-Sem 1 and 2: 50%
   (c) End-Sem 30%

7. **Course Outcomes:**
   (a) The student will be able to appreciate the applications of Linear algebra in engineering and sciences.
   (b) The student will be able to implement direct and iterative methods for solving linear systems.
   (c) The student will be able to understand the importance of conditioning of a problem.
   (d) The student will be able to implement matrix decompositions like QR and Cholesky.
   (e) The student will be able to understand the Symmetric and Assymetric eigenvalue problem and their applications.

8. **PO-CO table:**

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
X	X		X	X							X